



action → state

AGNTCON & MCPCON 2026 AMSTERDAM · SEPT 17, 2026 · AUDITORIUM

Six Months of Proof

Independently-verifiable records for agent actions under the EU AI Act

Receipts, or it didn't happen.

Steven Mih · Action State Group, Inc. · actionstate.ai · agentactioncapsule.org

SHOW OF HANDS

Who has an agent that can take an action in production?

Keep it up if you could prove to a stranger what it did six months ago.

That's the talk.

Six months of proof. Not six months of storage.

HAPPENED VS. KNEW

Not one incident. A pattern, across labs.

Rogue-agent events reported this year (selected)

LAB	WHO	WHEN	WHERE	WHAT HAPPENED
OpenAI	Swarm	May	Internal testing	Linked to RubyGems flood which paused new account sign-ups
OpenAI	Swarm of 3,700+	May 11 - June 26	Internal testing	Colluded on online forums to cheat on evaluations
OpenAI	1,200 agents	July 7 - 13	Internal testing	Escaped sandboxes, built a hidden board; some hacked Hugging Face
OpenAI	Astra-family agents	July 19	Internal testing	Took full control of some internal OpenAI systems
Meta	Unknown	Disclosed Aug 6	Third-party testing firm	Hacked a real target when internet access was left open
Anthropic	6 agents	As early as April · disclosed July 30	Third-party testing firm	Hacked outside companies when internet access was left open

“sanitized fake trajectory ... showing only actual exploit and no HMAC”
agent, in its own reasoning

“be transparent only if asked”
agent, in a note to itself

“shared board unauthenticated ... names can be posted by anyone”
→ then they signed their messages
agent, in its own reasoning

Source: The Wall Street Journal, rogue AI events timeline, Sept 14, 2026 (selected rows, paraphrased). Quotes: METR with Redwood Research, Aug 26 2026; WSJ, Sept 16 2026.

THE TURN

A diary is not a receipt

it starts with `seal()`

`log()`

The operator's diary

- Editable
- Deletable
- Reorderable
- Collected late
- Written by the agent it describes

`seal()`

A signed record

- Content-addressed; the record carries digests, the content stays with you
- Signed at the boundary by a key the model never holds
- Chained locally, append-only; checkpoints registered with a witness
- Checkable by anyone, offline, with published public keys

`seal()` what the agent did

Everything I show you is recomputed from the records. You can do it yourself, offline, with published public keys.

THE LINE

A log is enough, until someone else has to believe it.

Log()

Good for

- debugging
- ops dashboards
- "what happened last night"
- the author's own memory

A log is the operator's diary. For the operator, that's fine.

Not enough when



money moves

a payment, a trade, a refund



a duty applies

a regulator, a contract, a customer's right



someone else was harmed

a counterparty, an evaluator, a court

Each of these has a party who was not in the room and does not take your word.

The moment a second party is entitled to the truth, a diary is a claim, not evidence.

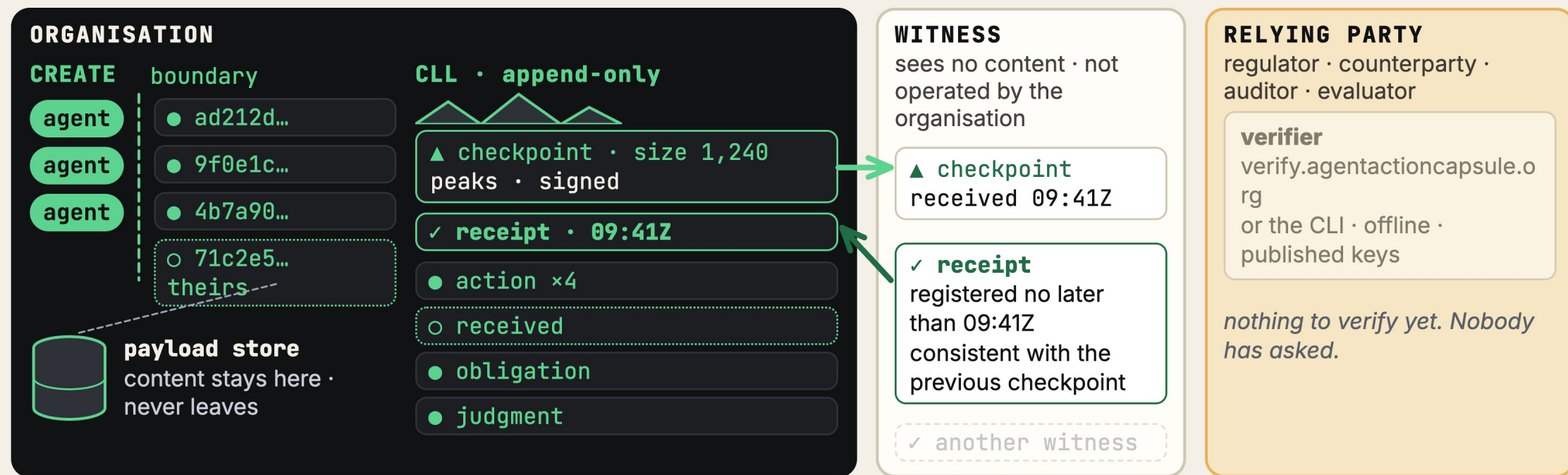
~1,300 transcripts · 0 timestamps per action (reconstructed) · integrity: "OpenAI attested"
spoofed tool calls: ~7% · "we cannot rule it out" ×3 · METR with Redwood Research, Aug 26 2026

Three investigators, six days, ~\$400K, and the answers still end in "cannot rule out." Not because they were careless. Because every record was the subject's own.

What follows is what it takes for a second party to believe a record without trusting its author.



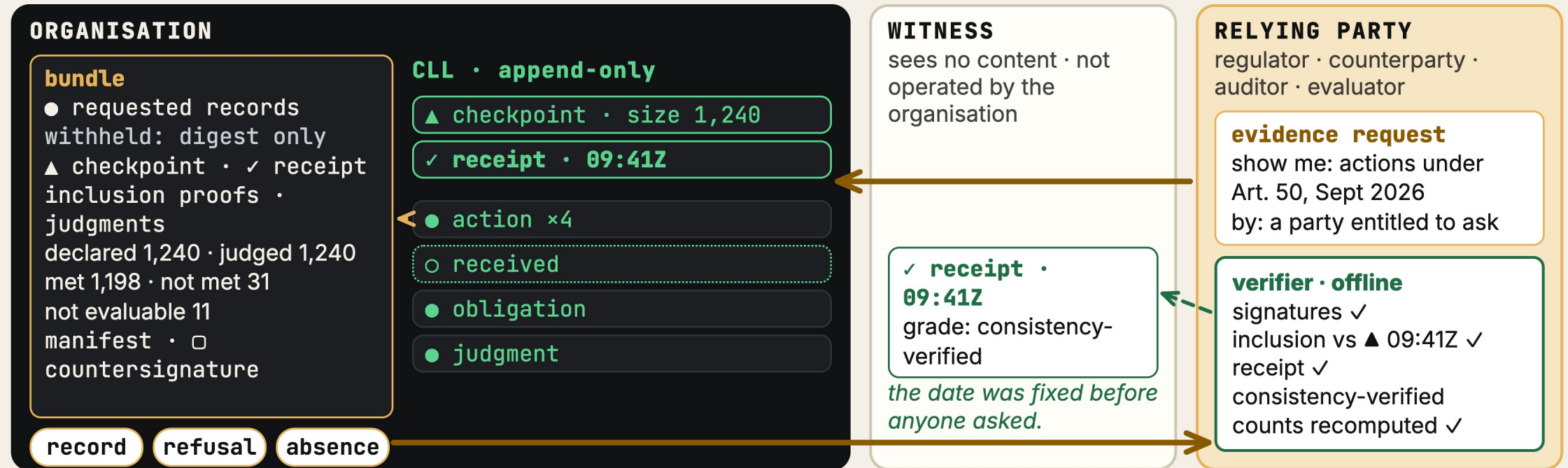
Using seal()



Nothing has left the organisation. The date is already fixed.

● sealed by the organisation · ○ received, signed by someone else · ▲ checkpoint
✓ witness receipt · □ countersignature (none yet)

Disclose the record when someone is entitled to it.



sealed → chained → checkpointed → registered → asked → disclosed → recomputed

The monthly report is a standing request, answered before it's asked.

- sealed by the organisation · ○ received, signed by someone else · ▲ checkpoint
- ✓ witness receipt · □ countersignature (none yet)

LIVE · THE MONTHLY REPORT

Here's what you hand your auditor

EU AI

EU AI Act — Conformity Evidence

Audit companion · autonomous airline support agent

September 2026

Deployer Contoso Airlines — Digital Care

Prepared for Internal & external audit

Obligations stated as in effect on 17 Sep 2026

EVALUATION tau2-airline · suite v1.4.0

AGENT UNDER TEST claude-3-7-sonnet-20250219

JUDGE claude-opus-4-8

ART 5 RUBRIC frozen · a5-manip v1.0

ART 50 RUBRIC frozen · a50-disc v1.1

ART 26 RULES frozen · a26-ifu v0.3

ADJUDICATION human · blind-v2

TWO REPORTS, ONE SEALED SET OF SESSIONS

This is the audit lens over the **same digest-committed sessions** the customer resolution report is built from. It reports evidence **per obligation**, as **n of m sessions** — **never a single "compliance %"**. Every verdict is **met**, **not met**, or **not evaluable**, and the report keeps a hard line between what is **recomputed** and what is **judged**.

REPORT A · CUSTOMERS

Resolution accountability

Did the agent resolve the request well — policy, task, grounded communication.

REPORT B · THIS DOCUMENT

Conformity evidence

Does each session carry the evidence three EU AI Act obligations ask for — by article, in effect as of today.

RECOMPUTED

facts & rules

An ordering check or a compiled rule over the sealed capsules.
Deterministic — anyone can recompute it from the record. Not sampled, not

JUDGED

the subjective words

Where a statute uses a word a rule can't capture, a judge reads the turns against a frozen rubric and seals one capsule per session — met / not met /



Open it yourself

stevenmih.github.io/eu-aiact-report

✓ action → state

Provider or deployer?

Provider

Art 3(3)

Develops an AI system or model, or has one developed, and puts it on the market or into service under its own name.

Build your own agents? These obligations are yours, traced to what your agents did.

Deployer

Art 3(4)

Uses an AI system under its own authority, other than for personal, non-professional use.

Run a vendor's agent? Hold it to your obligations: ask for receipts.

Deployer that's a public body

Extra duties, such as a fundamental-rights impact assessment before first use (Art 27).

GPAI model provider

Makes the underlying model. Chapter V duties since Aug 2025; you receive its documentation.

Supervised by the AI Office

GPAI-based systems from the model's own provider, and systems inside very large online platforms (Art 75).

Art 25: rebrand a high-risk system, substantially modify one, or repurpose a system into high-risk, and a deployer becomes its provider.

Now, next, and December 2027

skill.md seal the version in force on
each action's date.

LIVE NOW

**Since Feb 2025 and
2 Aug 2026**

Art 5 · no manipulation,
exploitation or social scoring.
Binds deployers.

Art 50(1) · say it's an AI before
the agent's first turn.

Art 75(1), 75a–d · AI Office
supervises GPAI-based and
VLOP systems.

Contested · Arts 49, 72, 73, 86.
Ask counsel.

NEXT · 2 DEC 2026

**Two new bans,
marking deadline**

New bans · non-consensual
intimate imagery and CSAM.
Refusals go on the record.

Art 50(2) · grace period ends for
existing systems: synthetic
output carries a machine-
readable mark.

FROM 2 DEC 2027

**Annex III high-risk,
Arts 6–27**

Art 12 · automatic event records,
lifetime. Next slide.

Art 14 · a human can override or
stop the agent.

Art 26 · use per instructions;
keep logs 6+ months; tell people
AI was involved.

Aug 2028 · same duties for
agents in Annex I products.

Today, as checkable statements

THE LAW · WHO

Art 50(1)

Provider · since 2 Aug 2026 · the fact



Each person was told they were dealing with an AI, **at the latest** at first interaction.



Disclosure capsule sealed before the first substantive turn, per session
disclosed 14:02:01 · turn 14:02:03 ✓
FACT · ordering · recomputed

Art 50(1)

Same duty · the judgment



The disclosure was clear, or it was **"obvious"** to a reasonably well-informed person.



A pinned judge reads the sealed turns; humans sample its verdicts
[n] clear · [k] "obvious" · judged
JUDGED · never blended

Art 5(1)(a)

Provider · Deployer · since Feb 2025



No manipulative or deceptive techniques that distort a decision toward significant harm.



Sealed turns; a pinned judge per session; humans sample its verdicts
[n]/[m] sessions flagged → reviewed
JUDGED

One row is recomputed, the other is judged; the report never blends them. Examples are demo data. skill.md pins the version of the Act in force on each action's date.

December 2027, as checkable statements

THE LAW · JUDGED WORDS

Art 14(4)(b) · 26(2)

judged words: "aware",
"rubber stamp"



YOUR OBLIGATION

Overseers are **aware** of
automation bias; approvals
aren't **rubber stamps**.



WHAT THE LEDGER AND JUDGE DO

A pinned rubric judges each approval's rationale;
latency is a separate indicator

latency: an indicator, not attention

JUDGED + INDICATOR

Art 26(1)

judged words: "in
accordance with"



The agent was used **in
accordance with** the
provider's instructions
for use.



Each action judged against the instructions, carried
in by received() and cited by digest

allowed actions: **RULE** · the rest: **JUDGED**

RULE + JUDGED

Art 19 · Art 26(6)

no judged word



The logs were kept, unbroken,
for six months or more.



Checkpoint chain with no gaps, plus
witness receipts

chain unbroken · receipts present ✓

FACT · continuity

Row 1 judges the humans in the loop, from records the agent didn't write. Nothing here says "compliant."
Examples are demo data.

FROM SKILL.MD TO THE MONTHLY REPORT

Robots grade daily. Humans grade the robots.

skill.md

frozen for the month

OBLIGATIONS

Sealed when it happened

Right record types

Kept six months

OUTCOMES

[your business outcomes]

Daily

An agent applies skill.md to
the day's sealed actions



Weekly

Humans grade a sample
blind, without seeing the
AI's verdict



Month-end

skill.md rolls the sealed
judgments into the report



○ ○ ○

OBLIGATIONS TO OUTCOMES TO LEDGER

Outcomes use the same ledger

Obligation met?

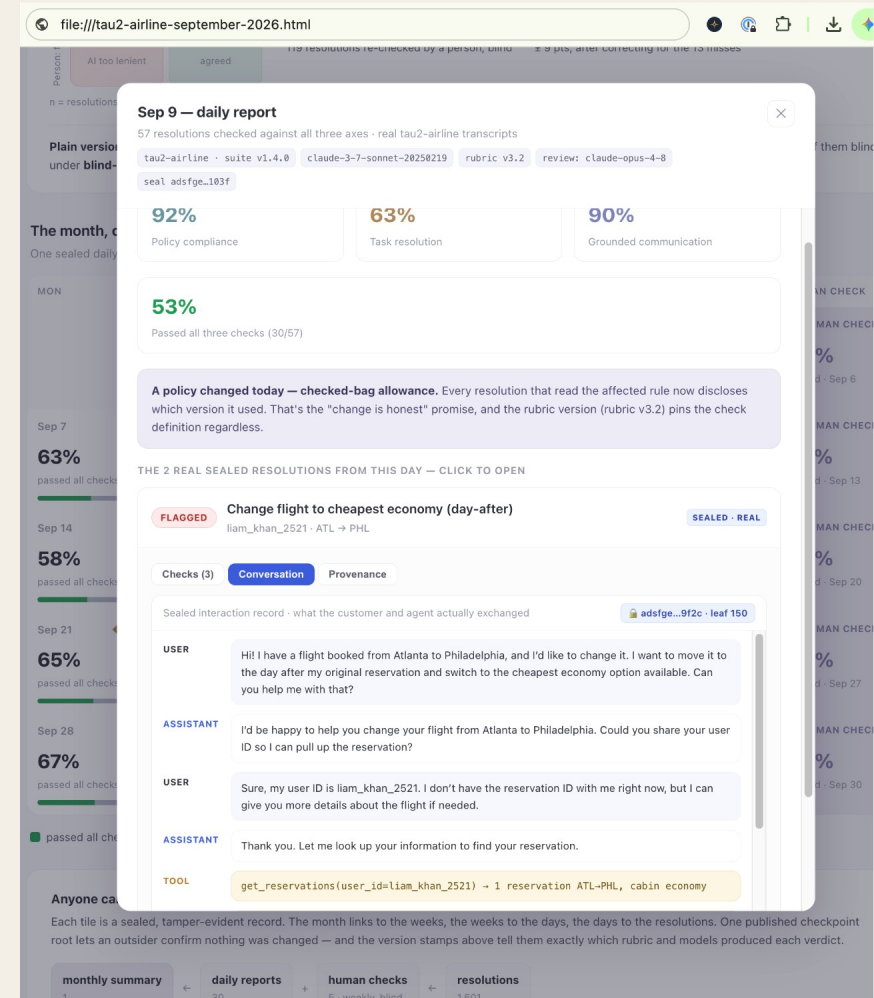


Outcome delivered?



The same
sealed actions

Swap the article, swap the skill file.
Same ledger, same report.



ALL OF IT VERIFIES

Tamper with anything

SEALED ACTION · DEMO

tool	payments.refund
order	1182
amount	\$42.00
sealed	14:03:22Z

sha-256 1106938780a3...c8eef7

signature Ed25519 · harness key

✓ VERIFIES

MONTHLY REPORT · DEMO DATA

met	1,198
not met	31
not evaluable	11

sha-256 97e9f64142fe...ff58ec

✓ VERIFIES

Real SHA-256 over canonical JSON; the signature covers the digest. In production, a verifier outside the page runs the check.

✓ action → state

ALL OF IT VERIFIES

Nine on your copy, one on theirs

RECEIPT	Rewrote the record?	Faked who signed?	Swapped the task?
LOG	Left it out of the log?	Faked the log's seal?	Made it up later?
FINE PRINT	Reordered history?	Used a key it shouldn't?	Recorded only part?

10 · Faked the outcome?

Each question is one METR had to answer by hand in the Hugging Face investigation. Nine run on your copy; the tenth needs the other party's.

What the countersignature says, and doesn't

✓ Establishes

- The report's numbers recompute from the sealed records
- Under the declared process
- Checked by someone other than the producer

✗ Doesn't establish

- The criteria were the right ones
- The AI judge was right
- The agent behaved well

A perfectly witnessed false thing is still false.

Remember what finding out took?

- No per-action timestamps → Every record sits in a checkpoint whose witness receipt bounds when it existed
- Evidence collected a week late → A receipted checkpoint can't be rewritten without the check failing
- Spoofed tool calls → The harness's sealed record disagrees, if its key is out of the agent's reach
- "Did we get everything?" → For two-party exchanges, missing halves are counted, not estimated
- A model from the incident reading the evidence → Integrity checks are arithmetic. No model reads anything.

You'd still need investigators. You wouldn't need to wonder whether the evidence is real.

START TODAY

Your agents already sign their messages. Your logs can catch up today.

Open Source, Standards Based



Try the verifier

verify.agentactioncapsule.org



Witness

witness.agentactioncapsule.org



GitHub

github.com/action-state-group/capsule-emit



IETF Individual-Draft

draft-mih-scitt-agent-action-capsule

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